

Router Configuration and Routing Protocols using Cisco Packet Tracer

Computer Networks Practical Assignment

Submitted By

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Introduction

This assignment demonstrates the configuration of Cisco routers using Cisco Packet Tracer. The practical activities include basic router configuration, implementation of static routing, and dynamic routing using RIP and EIGRP protocols. The objective is to establish successful communication among multiple networks and verify connectivity through routing tables and ping tests.

Task 1 – Basic Router Configuration

Objective

Configure the router hostname, secure console access with a password, and enable login authentication.

Configuration Commands

```
enable
configure terminal

hostname KandyNSBM_39271

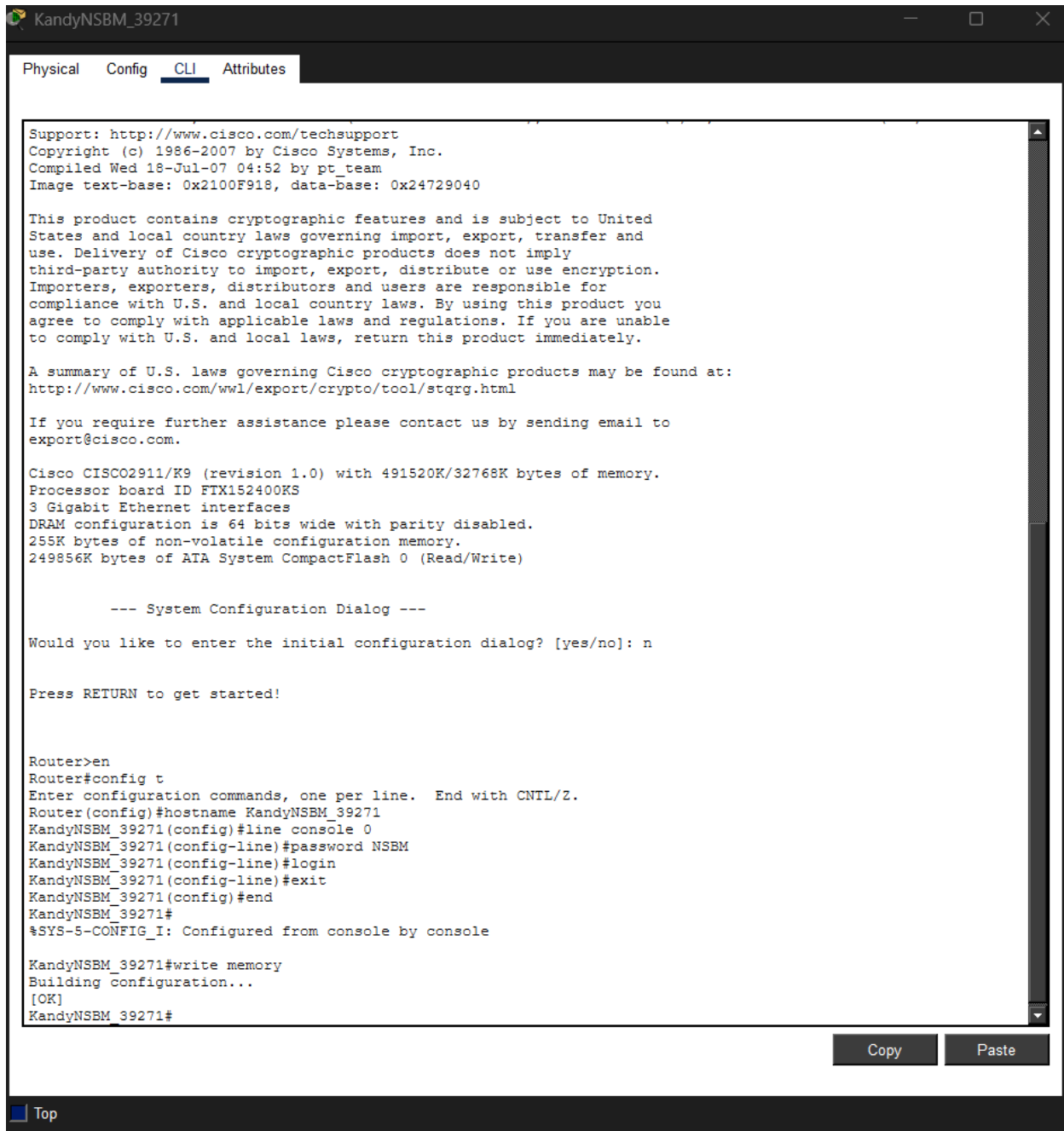
line console 0
password NSBM
login
```

```
end
write memory
```

Explanation of Commands

Command	Purpose
hostname KandyNSBM_39271	Changes the router hostname
line console 0	Accesses console configuration mode
password NSBM	Sets the console password
login	Enables password authentication
write memory	Saves configuration permanently

Hostname configuration and Console password configuration



```
KandyNSBM_39271
Physical Config CLI Attributes

Support: http://www.cisco.com/techsupport
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Compiled Wed 18-Jul-07 04:52 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
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A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname KandyNSBM_39271
KandyNSBM_39271(config)#line console 0
KandyNSBM_39271(config-line)#password NSBM
KandyNSBM_39271(config-line)#login
KandyNSBM_39271(config-line)#exit
KandyNSBM_39271(config)#end
KandyNSBM_39271#
#SYS-5-CONFIG_I: Configured from console by console

KandyNSBM_39271#write memory
Building Configuration...
[OK]
KandyNSBM_39271#
```

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Result

The router hostname was successfully changed to **KandyNSBM_39271** and console access was secured using the password **NSBM**.

Task 2 – Static Routing Configuration

Network Topology

The network consists of three routers:

- ComputingRouter
- BusinessRouter
- ScienceRouter

Network Addressing

Device	Interface	IP Address
ComputingRouter	G0/2	192.168.1.1
ComputingRouter	G0/1	12.0.0.1
ComputingRouter	G0/0	11.0.0.1
BusinessRouter	G0/2	192.168.2.1
BusinessRouter	G0/1	12.0.0.2
BusinessRouter	G0/0	10.0.0.1
ScienceRouter	G0/1	192.168.3.1
ScienceRouter	G0/0	10.0.0.2
ScienceRouter	G0/2	11.0.0.2

Router Configurations

ComputingRouter

```
interface g0/2
ip address 192.168.1.1 255.255.255.0
no shutdown
```

```
interface g0/1
ip address 12.0.0.1 255.255.255.252
no shutdown
```

```
interface g0/0
ip address 11.0.0.1 255.255.255.252
no shutdown
```

```
ip route 192.168.2.0 255.255.255.0 12.0.0.2
ip route 192.168.3.0 255.255.255.0 11.0.0.2
```

BusinessRouter

```
interface g0/2
ip address 192.168.2.1 255.255.255.0
no shutdown
```

```
interface g0/0
ip address 10.0.0.1 255.255.255.252
no shutdown
```

```
interface g0/1
ip address 12.0.0.2 255.255.255.252
no shutdown
```

```
ip route 192.168.1.0 255.255.255.0 12.0.0.1
ip route 192.168.3.0 255.255.255.0 10.0.0.2
```

ScienceRouter

```
interface g0/1
ip address 192.168.3.1 255.255.255.0
no shutdown
```

```
interface g0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
```

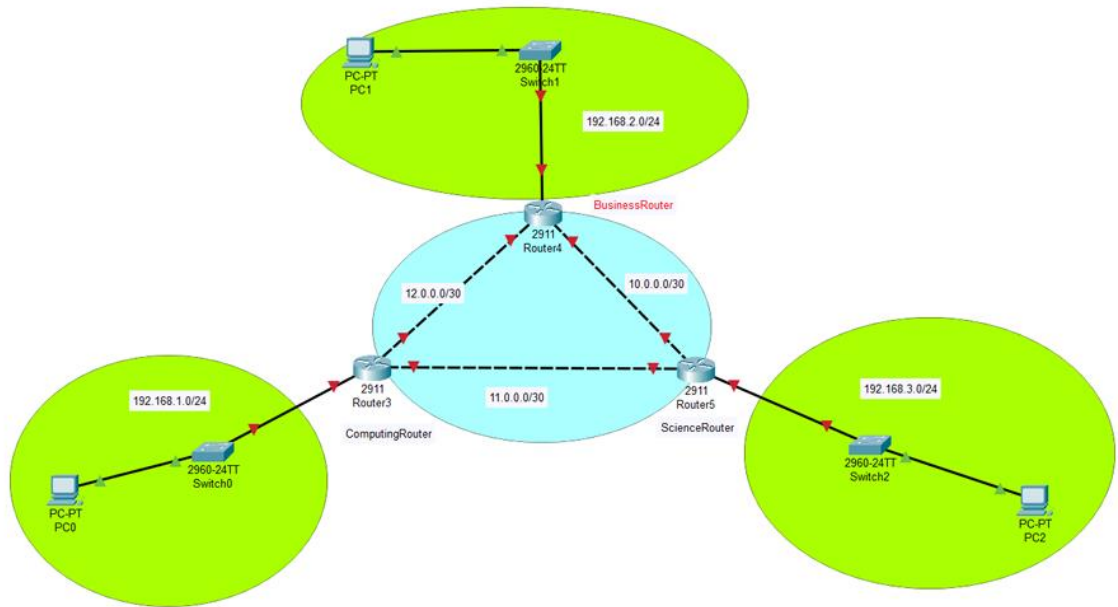
```
interface g0/2
ip address 11.0.0.2 255.255.255.252
no shutdown
```

```
ip route 192.168.1.0 255.255.255.0 11.0.0.1
ip route 192.168.2.0 255.255.255.0 10.0.0.1
```

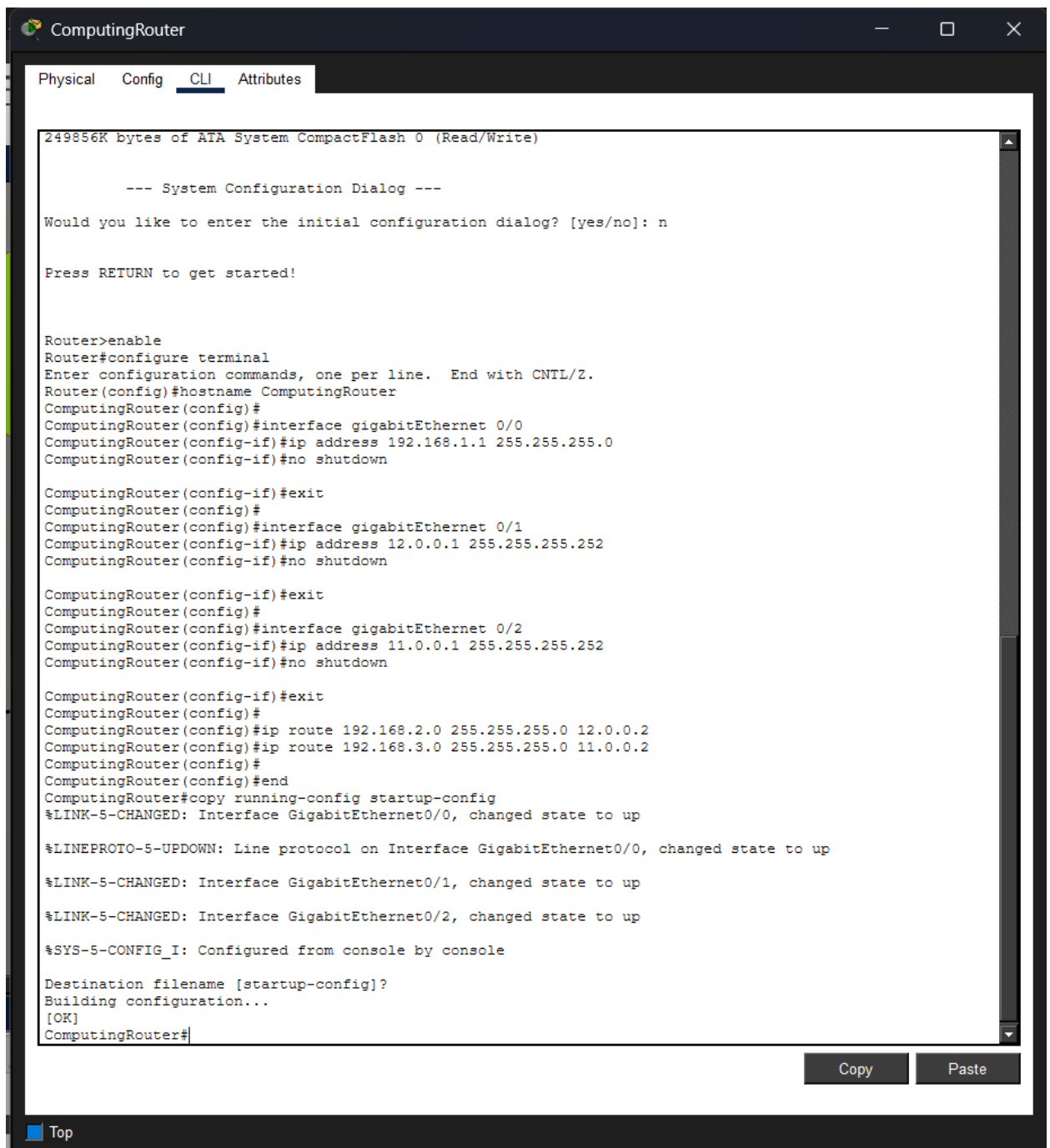
Explanation of Static Routing

Static routing manually defines routes to remote networks. Each router is configured with destination networks and the next-hop address to reach those networks. This allows routers to forward packets correctly between different LANs.

Network topology diagram



ComputingRouter configuration



ComputingRouter

Physical Config CLI Attributes

249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ComputingRouter
ComputingRouter(config)#
ComputingRouter(config)#interface gigabitEthernet 0/0
ComputingRouter(config-if)#ip address 192.168.1.1 255.255.255.0
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#exit
ComputingRouter(config)#
ComputingRouter(config)#interface gigabitEthernet 0/1
ComputingRouter(config-if)#ip address 12.0.0.1 255.255.255.252
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#exit
ComputingRouter(config)#
ComputingRouter(config)#interface gigabitEthernet 0/2
ComputingRouter(config-if)#ip address 11.0.0.1 255.255.255.252
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#exit
ComputingRouter(config)#
ComputingRouter(config)#ip route 192.168.2.0 255.255.255.0 12.0.0.2
ComputingRouter(config)#ip route 192.168.3.0 255.255.255.0 11.0.0.2
ComputingRouter(config)#
ComputingRouter(config)#end
ComputingRouter#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ComputingRouter#
```

Copy Paste

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BusinessRouter configuration



The screenshot shows a terminal window titled "BusinessRouter" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a "System Configuration Dialog". The dialog asks if the user wants to enter the initial configuration, which is declined with 'n'. The user then enters 'enable' to access privileged mode and 'configure terminal' to enter configuration mode. Three interfaces (0/2, 0/0, and 0/1) are configured with IP addresses and no shutdown. Static routes are added for the 192.168.1.0 and 192.168.3.0 networks. The running configuration is copied to the startup configuration, and the system is rebooted. The final prompt is "BusinessRouter#".

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname BusinessRouter
BusinessRouter(config)#
BusinessRouter(config)#interface gigabitEthernet 0/2
BusinessRouter(config-if)#ip address 192.168.2.1 255.255.255.0
BusinessRouter(config-if)#no shutdown

BusinessRouter(config-if)#exit
BusinessRouter(config)#
BusinessRouter(config)#interface gigabitEthernet 0/0
BusinessRouter(config-if)#ip address 10.0.0.1 255.255.255.252
BusinessRouter(config-if)#no shutdown

BusinessRouter(config-if)#exit
BusinessRouter(config)#
BusinessRouter(config)#interface gigabitEthernet 0/1
BusinessRouter(config-if)#ip address 12.0.0.2 255.255.255.252
BusinessRouter(config-if)#no shutdown

BusinessRouter(config-if)#exit
BusinessRouter(config)#
BusinessRouter(config)#ip route 192.168.1.0 255.255.255.0 12.0.0.1
BusinessRouter(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
BusinessRouter(config)#
BusinessRouter(config)#end
BusinessRouter#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

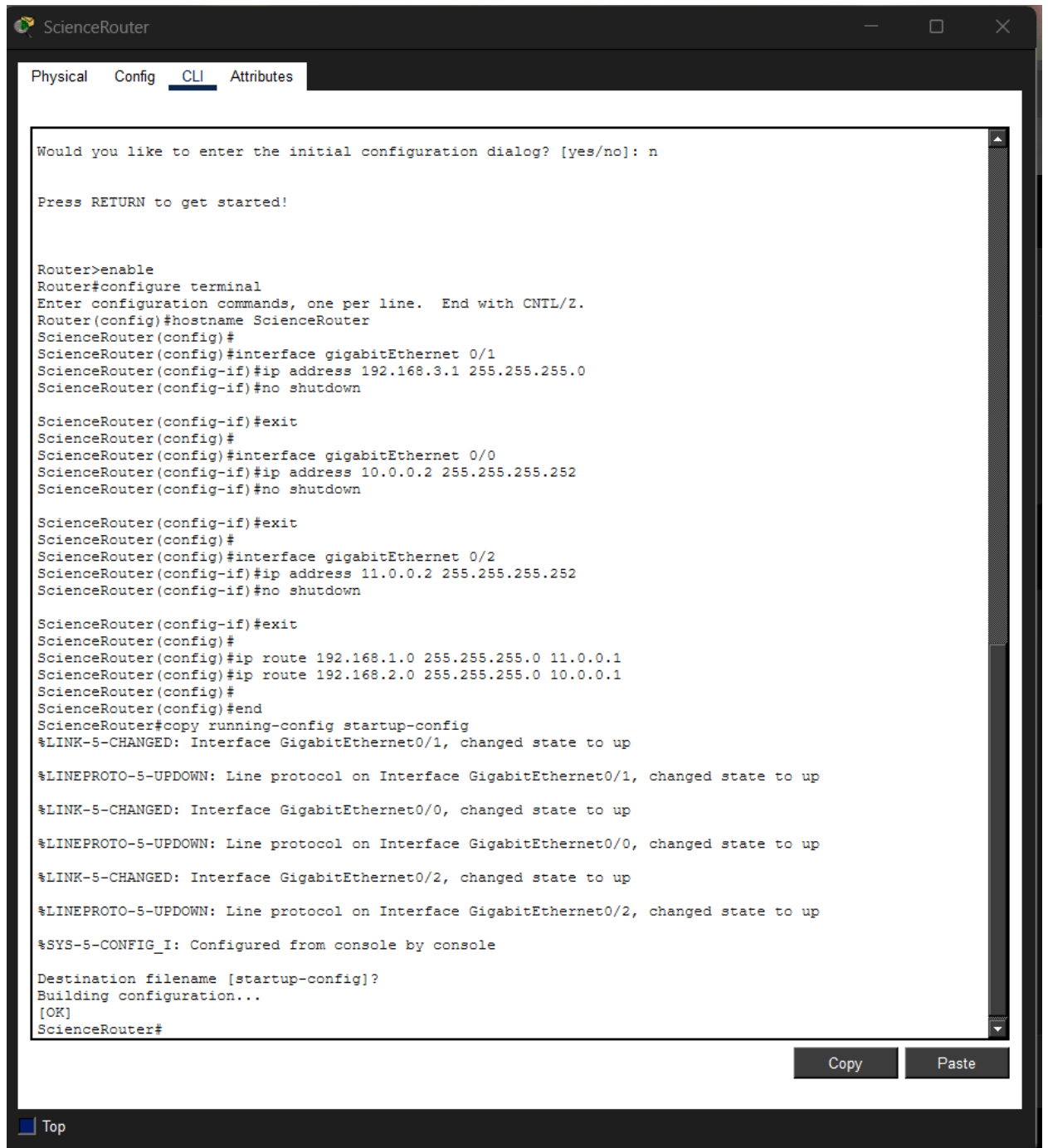
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
BusinessRouter#
```

Copy Paste

Top

ScienceRouter configuration



The screenshot shows a terminal window titled "ScienceRouter" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of commands and their outputs. The process begins with a prompt to enter the initial configuration dialog, followed by enabling the terminal and configuring the router's hostname to "ScienceRouter". Three interfaces (0/1, 0/0, and 0/2) are configured with specific IP addresses and no shutdown. Static routes are then added for each interface. Finally, the running configuration is saved to the startup configuration, and the system status is confirmed. The terminal ends with a "Top" button and "Copy" and "Paste" buttons.

```
ScienceRouter
Physical Config CLI Attributes

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ScienceRouter
ScienceRouter(config)#
ScienceRouter(config)#interface gigabitEthernet 0/1
ScienceRouter(config-if)#ip address 192.168.3.1 255.255.255.0
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#exit
ScienceRouter(config)#
ScienceRouter(config)#interface gigabitEthernet 0/0
ScienceRouter(config-if)#ip address 10.0.0.2 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#exit
ScienceRouter(config)#
ScienceRouter(config)#interface gigabitEthernet 0/2
ScienceRouter(config-if)#ip address 11.0.0.2 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#exit
ScienceRouter(config)#
ScienceRouter(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.1
ScienceRouter(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
ScienceRouter(config)#
ScienceRouter(config)#end
ScienceRouter#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ScienceRouter#
```

Top Copy Paste

ComputingRouter routing table

ComputingRouter

Physical Config **CLI** Attributes

```
ComputingRouter(config)#ip route 192.168.2.0 255.255.255.0 12.0.0.2
ComputingRouter(config)#ip route 192.168.3.0 255.255.255.0 11.0.0.2
ComputingRouter(config)#
ComputingRouter(config)#end
ComputingRouter#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ComputingRouter#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

ComputingRouter#
ComputingRouter#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1      YES manual up          up
GigabitEthernet0/1  12.0.0.1         YES manual up          up
GigabitEthernet0/2  11.0.0.1         YES manual up          up
Vlan1          unassigned       YES unset  administratively down down
ComputingRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, GigabitEthernet0/2
L       11.0.0.1/32 is directly connected, GigabitEthernet0/2
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, GigabitEthernet0/1
L       12.0.0.1/32 is directly connected, GigabitEthernet0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S       192.168.2.0/24 [1/0] via 12.0.0.2
S       192.168.3.0/24 [1/0] via 11.0.0.2

ComputingRouter#
```

Copy Paste

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BusinessRouter routing table

BusinessRouter
Physical Config **CLI** Attributes

```

BusinessRouter(config)#ip route 192.168.1.0 255.255.255.0 12.0.0.1
BusinessRouter(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
BusinessRouter(config)#
BusinessRouter(config)#end
BusinessRouter#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
BusinessRouter#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

BusinessRouter#
BusinessRouter#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.0.1	YES	manual	up	up
GigabitEthernet0/1	12.0.0.2	YES	manual	up	up
GigabitEthernet0/2	192.168.2.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```

BusinessRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, GigabitEthernet0/1
L       12.0.0.2/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 12.0.0.1
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/2
L       192.168.2.1/32 is directly connected, GigabitEthernet0/2
S       192.168.3.0/24 [1/0] via 10.0.0.2

BusinessRouter#

```

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ScienceRouter routing table

ScienceRouter

Physical Config CLI Attributes

```
ScienceRouter(config-if)#exit
ScienceRouter(config)#
ScienceRouter(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.1
ScienceRouter(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
ScienceRouter(config)#
ScienceRouter(config)#end
ScienceRouter#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ScienceRouter#show ip interface brief


| Interface          | IP-Address  | OK? | Method | Status                | Protocol |
|--------------------|-------------|-----|--------|-----------------------|----------|
| GigabitEthernet0/0 | 10.0.0.2    | YES | manual | up                    | up       |
| GigabitEthernet0/1 | 192.168.3.1 | YES | manual | up                    | up       |
| GigabitEthernet0/2 | 11.0.0.2    | YES | manual | up                    | up       |
| Vlan1              | unassigned  | YES | unset  | administratively down | down     |


ScienceRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
    C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
    L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
    C       11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
    C       11.0.0.0/30 is directly connected, GigabitEthernet0/2
    L       11.0.0.2/32 is directly connected, GigabitEthernet0/2
    S       192.168.1.0/24 [1/0] via 11.0.0.1
    S       192.168.2.0/24 [1/0] via 10.0.0.1
    C       192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
    C       192.168.3.0/24 is directly connected, GigabitEthernet0/1
    L       192.168.3.1/32 is directly connected, GigabitEthernet0/1

ScienceRouter#
```

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Ping Test Network A ↔ Network B

```
PC0(2)
Physical Config Desktop Programming Attributes
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.10

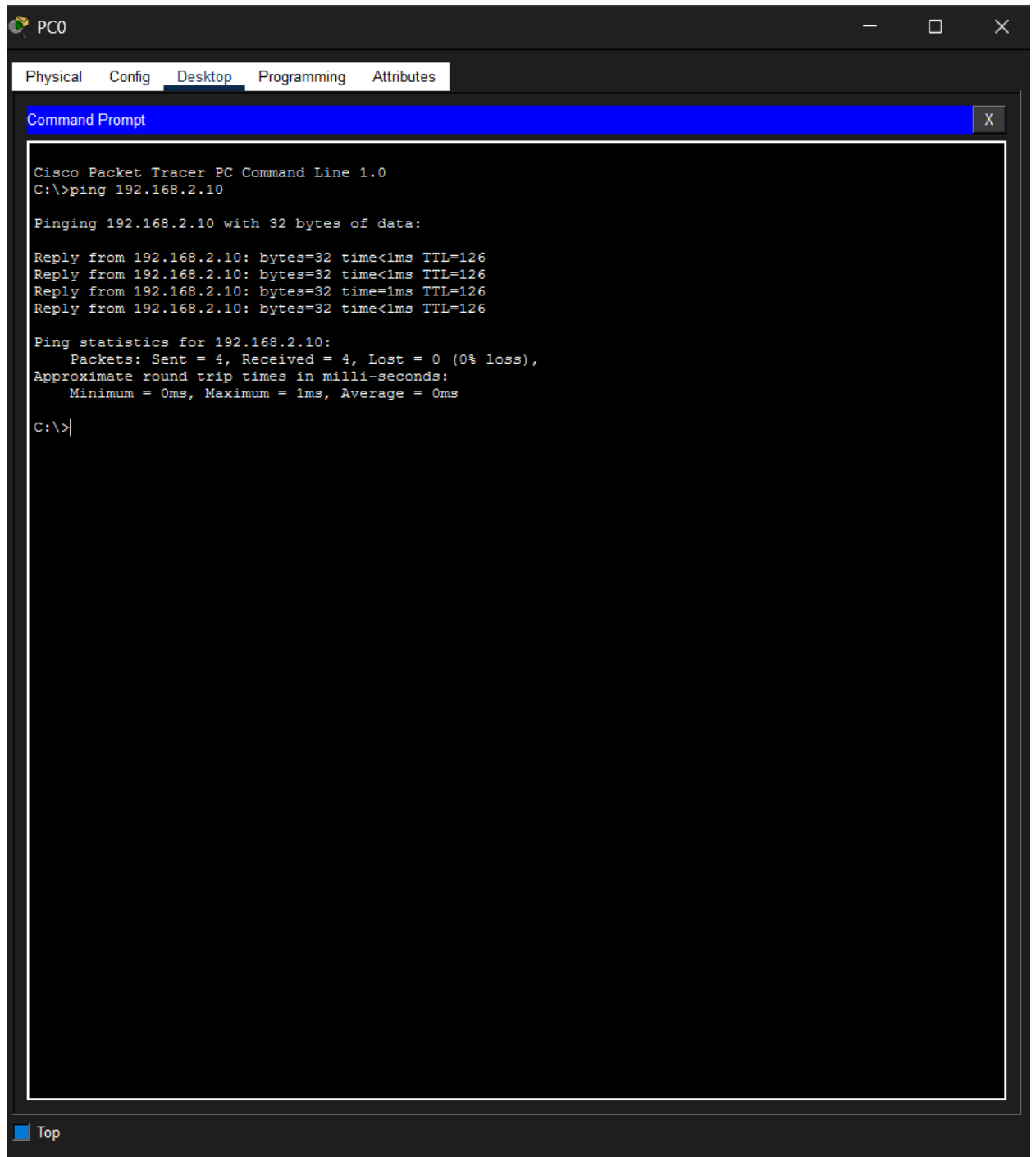
Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126
Reply from 192.168.1.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Ping Test Network A ↔ Network C



The screenshot shows a PC window titled "PC0" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a "Command Prompt" window. The command prompt shows the execution of a ping command to 192.168.2.10, resulting in four successful replies with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

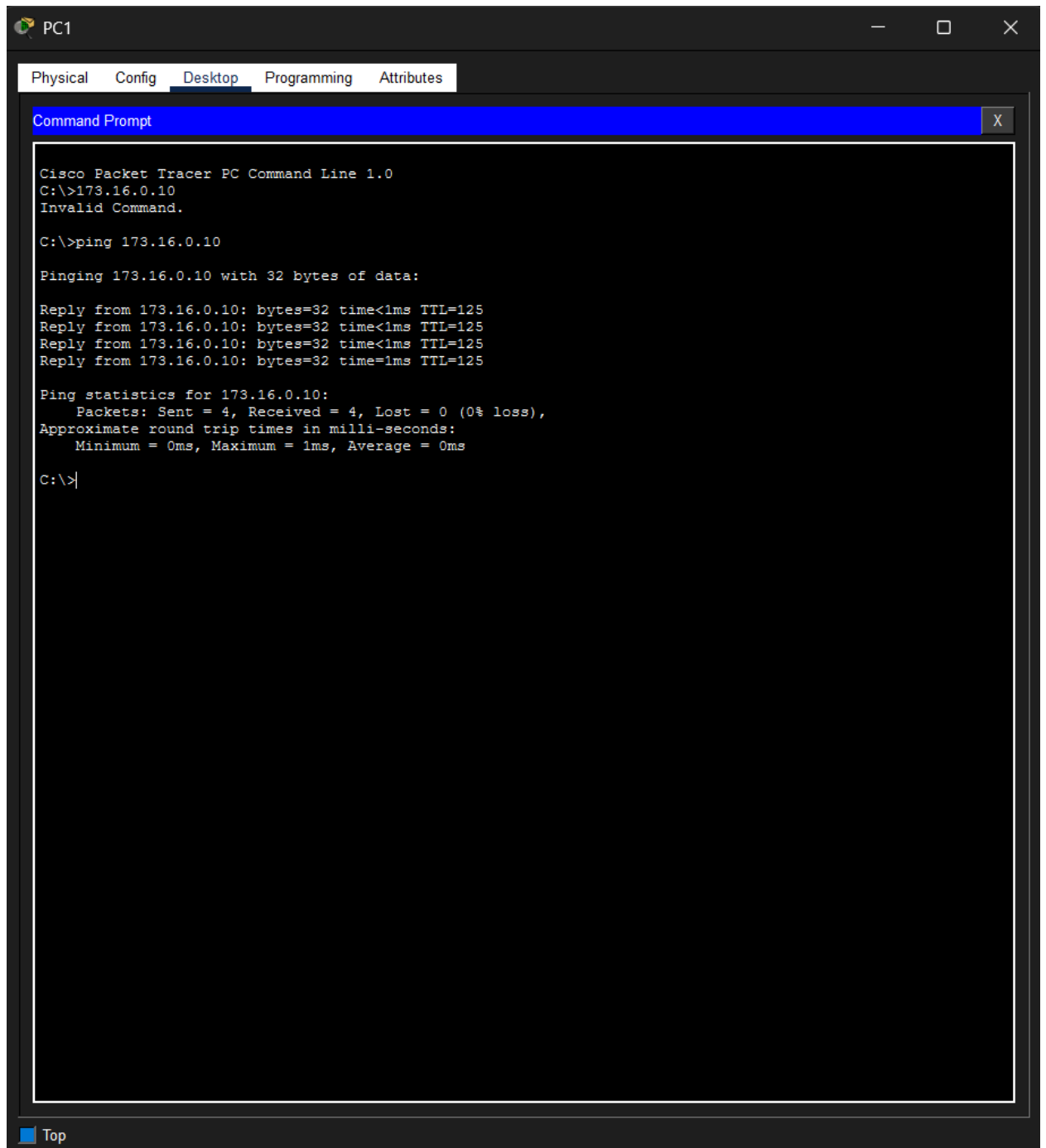
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

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Ping Test Network B ↔ Network C



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC1. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>173.16.0.10
Invalid Command.

C:\>ping 173.16.0.10

Pinging 173.16.0.10 with 32 bytes of data:

Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time=1ms TTL=125

Ping statistics for 173.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

At the bottom left of the Command Prompt window, there is a "Top" button.

Results

All routers successfully exchanged packets using static routes. Ping tests confirmed successful communication between all networks.

Task 3 – Dynamic Routing Configuration

Part A – RIP Routing

Objective

Configure Routing Information Protocol (RIP Version 2) to automatically exchange routing information among routers.

RIP Configuration

KandyNSBM Router

```
router rip
version 2
no auto-summary
network 14.0.0.0
network 172.16.0.0
```

ColomboNSBM Router

```
router rip
version 2
no auto-summary
network 14.0.0.0
network 15.0.0.0
```

GalleNSBM Router

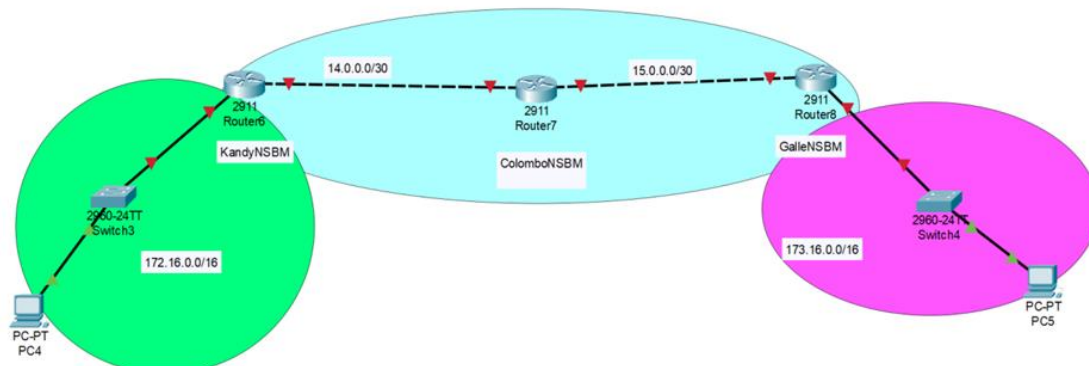
```
router rip
version 2
no auto-summary
network 15.0.0.0
network 173.16.0.0
```

Explanation

RIP is a distance-vector routing protocol that exchanges routing information periodically. Version 2 supports subnet masks and classless routing.

RIP Network Diagram

Task 3 – Dynamic Routing Configuration (15 Marks)



KandyNSBM RIP Configuration

The screenshot shows a terminal window titled "KandyNSBM" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of commands and their outputs. The configuration process starts with enabling the terminal, setting the hostname to "KandyNSBM", and configuring two interfaces: "gigabitEthernet 0/0" with IP "14.0.0.1" and "gigabitEthernet 0/1" with IP "172.16.0.1". After saving the configuration, the user enters the "router rip" configuration mode, sets the version to 2, disables auto-summary, and advertises the two networks. The process concludes with saving the configuration again and returning to the CLI prompt.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname KandyNSBM
KandyNSBM(config)#
KandyNSBM(config)#interface gigabitEthernet 0/0
KandyNSBM(config-if)#ip address 14.0.0.1 255.255.255.252
KandyNSBM(config-if)#no shutdown

KandyNSBM(config-if)#exit
KandyNSBM(config)#
KandyNSBM(config)#interface gigabitEthernet 0/1
KandyNSBM(config-if)#ip address 172.16.0.1 255.255.0.0
KandyNSBM(config-if)#no shutdown

KandyNSBM(config-if)#exit
KandyNSBM(config)#
KandyNSBM(config)#end
KandyNSBM#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
KandyNSBM#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

KandyNSBM#
KandyNSBM#enable
KandyNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
KandyNSBM(config)#
KandyNSBM(config)#router rip
KandyNSBM(config-router)#version 2
KandyNSBM(config-router)#no auto-summary
KandyNSBM(config-router)#network 14.0.0.0
KandyNSBM(config-router)#network 172.16.0.0
KandyNSBM(config-router)#exit
KandyNSBM(config)#
KandyNSBM(config)#end
KandyNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
KandyNSBM#
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

ColomboNSBM RIP Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ColomboNSBM
ColomboNSBM(config)#
ColomboNSBM(config)#interface gigabitEthernet 0/0
ColomboNSBM(config-if)#ip address 14.0.0.2 255.255.255.252
ColomboNSBM(config-if)#no shutdown

ColomboNSBM(config-if)#exit
ColomboNSBM(config)#
ColomboNSBM(config)#interface gigabitEthernet 0/1
ColomboNSBM(config-if)#ip address 15.0.0.1 255.255.255.252
ColomboNSBM(config-if)#no shutdown

ColomboNSBM(config-if)#exit
ColomboNSBM(config)#
ColomboNSBM(config)#end
ColomboNSBM#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ColomboNSBM#
ColomboNSBM#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

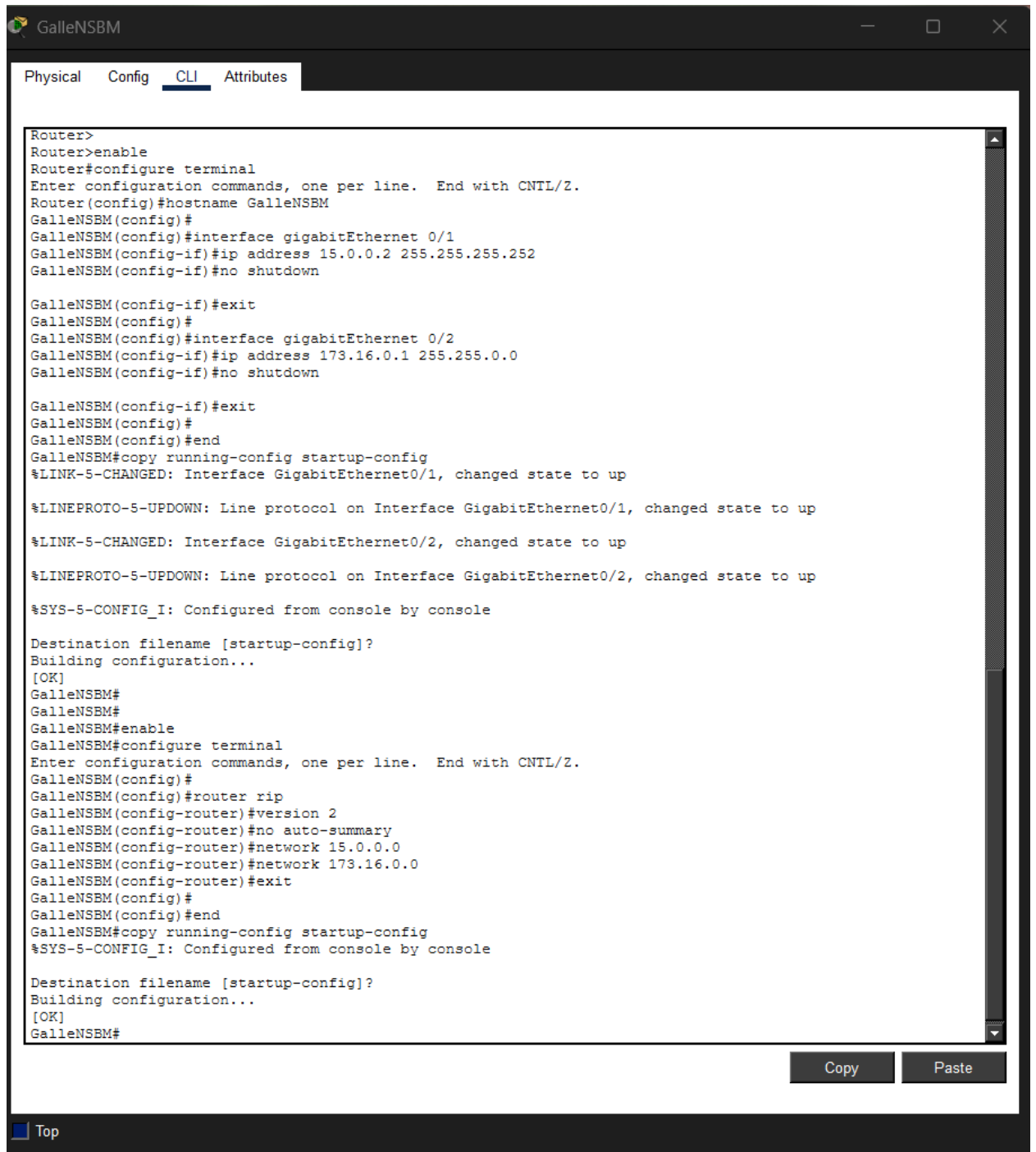
ColomboNSBM#
ColomboNSBM#enable
ColomboNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ColomboNSBM(config)#
ColomboNSBM(config)#router rip
ColomboNSBM(config-router)#version 2
ColomboNSBM(config-router)#no auto-summary
ColomboNSBM(config-router)#network 14.0.0.0
ColomboNSBM(config-router)#network 15.0.0.0
ColomboNSBM(config-router)#exit
ColomboNSBM(config)#
ColomboNSBM(config)#end
ColomboNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ColomboNSBM#
```

Copy Paste

Top

GalleNSBM RIP Configuration



The screenshot shows a terminal window titled "GalleNSBM" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of commands and their outputs. The commands configure two interfaces (0/1 and 0/2) with IP addresses, enable them, and copy the running configuration to the startup configuration. Status messages indicate that the interfaces are up and line protocols are up. The configuration is then saved to the startup configuration file.

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname GalleNSBM
GalleNSBM(config)#
GalleNSBM(config)#interface gigabitEthernet 0/1
GalleNSBM(config-if)#ip address 15.0.0.2 255.255.255.252
GalleNSBM(config-if)#no shutdown

GalleNSBM(config-if)#exit
GalleNSBM(config)#
GalleNSBM(config)#interface gigabitEthernet 0/2
GalleNSBM(config-if)#ip address 173.16.0.1 255.255.0.0
GalleNSBM(config-if)#no shutdown

GalleNSBM(config-if)#exit
GalleNSBM(config)#
GalleNSBM(config)#end
GalleNSBM#copy running-config startup-config
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

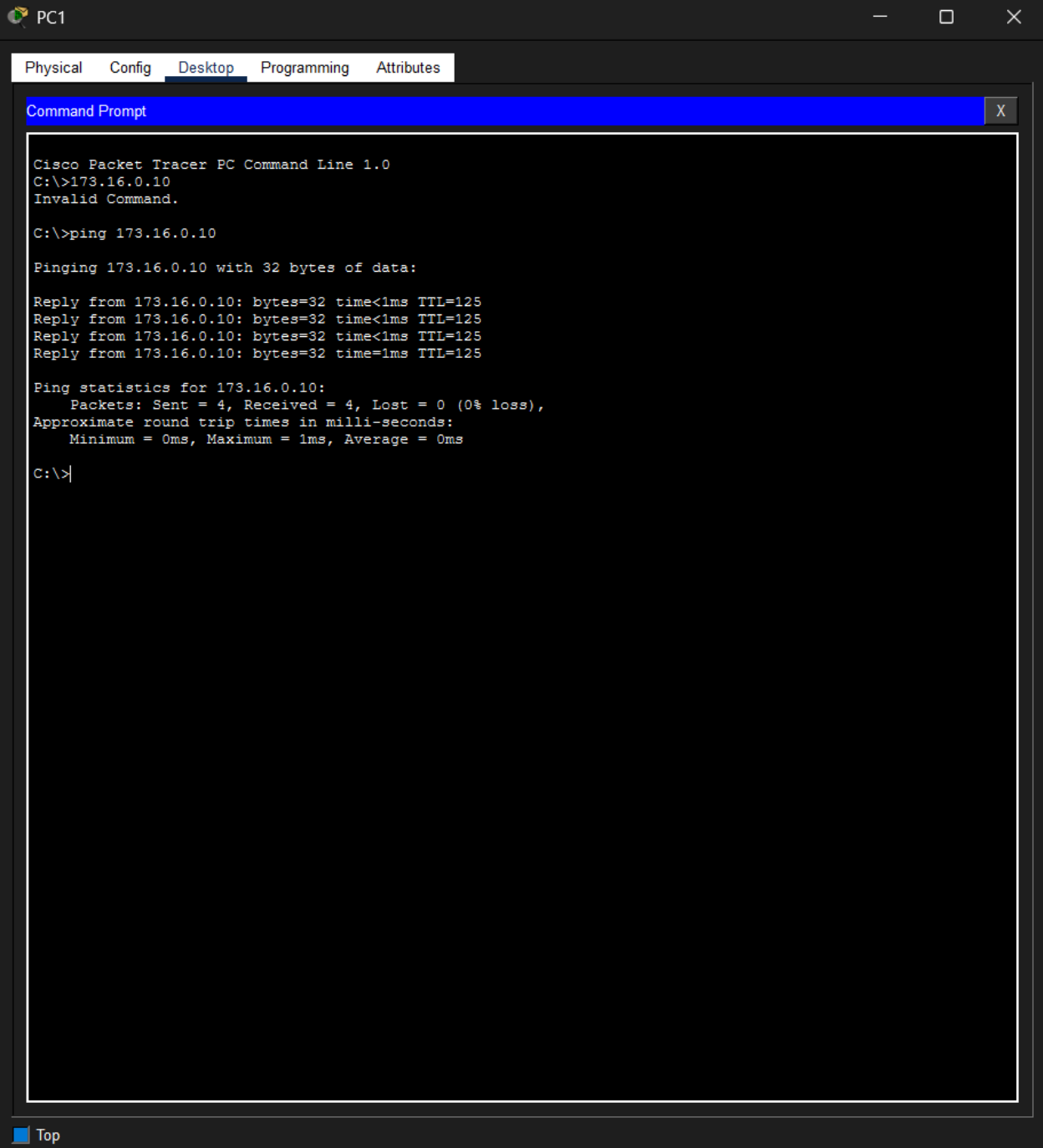
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
GalleNSBM#
GalleNSBM#
GalleNSBM#enable
GalleNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
GalleNSBM(config)#
GalleNSBM(config)#router rip
GalleNSBM(config-router)#version 2
GalleNSBM(config-router)#no auto-summary
GalleNSBM(config-router)#network 15.0.0.0
GalleNSBM(config-router)#network 173.16.0.0
GalleNSBM(config-router)#exit
GalleNSBM(config)#
GalleNSBM(config)#end
GalleNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
GalleNSBM#
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

Successful Ping Test



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>173.16.0.10
Invalid Command.

C:\>ping 173.16.0.10

Pinging 173.16.0.10 with 32 bytes of data:

Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time=1ms TTL=125

Ping statistics for 173.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

1. [Top](#)

Result

RIP successfully learned remote routes and enabled communication between all networks.

Part B – EIGRP Routing

Objective

Replace RIP with EIGRP and verify network connectivity.

KandyNSBM Router

```
no router rip
```

```
router eigrp 100  
no auto-summary  
network 14.0.0.0 0.0.0.3  
network 172.16.0.0 0.0.255.255
```

ColomboNSBM Router

```
no router rip
```

```
router eigrp 100  
no auto-summary  
network 14.0.0.0 0.0.0.3  
network 15.0.0.0 0.0.0.3
```

GalleNSBM Router

```
no router rip
```

```
router eigrp 100  
no auto-summary  
network 15.0.0.0 0.0.0.3  
network 173.16.0.0 0.0.255.255
```

Explanation

EIGRP is an advanced distance-vector routing protocol developed by Cisco. It converges faster than RIP and uses metrics based on bandwidth and delay to determine optimal routes.

KandyNSBM EIGRP Configuration

```
[OK]
KandyNSBM#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

KandyNSBM#
KandyNSBM#enable
KandyNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
KandyNSBM(config)#
KandyNSBM(config)#router 100
KandyNSBM(config-router)#version 2
KandyNSBM(config-router)#no auto-summary
KandyNSBM(config-router)#network 14.0.0.0
KandyNSBM(config-router)#network 172.16.0.0
KandyNSBM(config-router)#exit
KandyNSBM(config)#
KandyNSBM(config)#end
KandyNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
KandyNSBM#enable
KandyNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
KandyNSBM(config)#
KandyNSBM(config)#router eigrp 100
KandyNSBM(config-router)#no auto-summary
KandyNSBM(config-router)#network 14.0.0.0 0.0.0.3
KandyNSBM(config-router)#network 172.16.0.0 0.0.255.255
KandyNSBM(config-router)#exit
KandyNSBM(config)#
KandyNSBM(config)#end
KandyNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
KandyNSBM#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 14.0.0.2 (GigabitEthernet0/0) is up: new adjacency
```

Copy Paste

Top

ColomboNSBM EIGRP Configuration

```
[OK]
ColomboNSBM#
ColomboNSBM#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

ColomboNSBM#
ColomboNSBM#enable
ColomboNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ColomboNSBM(config)#
ColomboNSBM(config)#router rip
ColomboNSBM(config-router)#version 2
ColomboNSBM(config-router)#no auto-summary
ColomboNSBM(config-router)#network 14.0.0.0
ColomboNSBM(config-router)#network 15.0.0.0
ColomboNSBM(config-router)#exit
ColomboNSBM(config)#
ColomboNSBM(config)#end
ColomboNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ColomboNSBM#enable
ColomboNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ColomboNSBM(config)#no router rip
ColomboNSBM(config)#end
ColomboNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
ColomboNSBM#enable
ColomboNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ColomboNSBM(config)#
ColomboNSBM(config)#router eigrp 100
ColomboNSBM(config-router)#no auto-summary
ColomboNSBM(config-router)#network 14.0.0.0 0.0.0.3
ColomboNSBM(config-router)#network 15.0.0.0 0.0.0.3
ColomboNSBM(config-router)#exit
ColomboNSBM(config)#
ColomboNSBM(config)#end
ColomboNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 14.0.0.1 (GigabitEthernet0/0) is up: new adjacency

Destination filename [startup-config]?
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 15.0.0.2 (GigabitEthernet0/1) is up: new adjacency
```

Copy Paste

Top

GalleNSBM EIGRP Configuration

```
GalleNSBM
Physical Config CLI Attributes
Destination filename [startup-config]?
Building configuration...
[OK]
GalleNSBM#
GalleNSBM#enable
GalleNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
GalleNSBM(config)#
GalleNSBM(config)#router rip
GalleNSBM(config-router)#version 2
GalleNSBM(config-router)#no auto-summary
GalleNSBM(config-router)#network 15.0.0.0
GalleNSBM(config-router)#network 173.16.0.0
GalleNSBM(config-router)#exit
GalleNSBM(config)#
GalleNSBM(config)#end
GalleNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
GalleNSBM#enable
GalleNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
GalleNSBM(config)#no router rip
GalleNSBM(config)#end
GalleNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
GalleNSBM#enable
GalleNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
GalleNSBM(config)#
GalleNSBM(config)#router eigrp 100
GalleNSBM(config-router)#no auto-summary
GalleNSBM(config-router)#network 15.0.0.0 0.0.0.3
GalleNSBM(config-router)#network 173.16.0.0 0.0.255.255
GalleNSBM(config-router)#exit
GalleNSBM(config)#
GalleNSBM(config)#end
GalleNSBM#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 15.0.0.1 (GigabitEthernet0/1) is up: new adjacency

Destination filename [startup-config]?
Building configuration...
[OK]
GalleNSBM#
```

Copy Paste

Top

KandyNSBM Routing Table

```
KandyNSBM
KandyNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/0
L       14.0.0.1/32 is directly connected, GigabitEthernet0/0
D       15.0.0.0/30 is subnetted, 1 subnets
       15.0.0.0/30 [90/3072] via 14.0.0.2, 00:04:26, GigabitEthernet0/0
D       172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, GigabitEthernet0/1
L       172.16.0.1/32 is directly connected, GigabitEthernet0/1
D       173.16.0.0/16 [90/3328] via 14.0.0.2, 00:04:15, GigabitEthernet0/0

KandyNSBM#show ip protocols

Routing Protocol is "eigrp 100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
    EIGRP-IPv4 Protocol for AS(100)
      Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
      NSF-aware route hold timer is 240
      Router-ID: 14.0.0.1
      Topology : 0 (base)
        Active Timer: 3 min
        Distance: internal 90 external 170
        Maximum path: 4
        Maximum hopcount 100
        Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
    14.0.0.0/30
    172.16.0.0
  Routing Information Sources:
    Gateway         Distance         Last Update
    14.0.0.2         90               975457
  Distance: internal 90 external 170

KandyNSBM#
KandyNSBM#
```

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KandyNSBM EIGRP Neighbors

KandyNSBM

PhysicalConfigCLIAttributes

```

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   14.0.0.0/30 is directly connected, GigabitEthernet0/0
L   14.0.0.1/32 is directly connected, GigabitEthernet0/0
15.0.0.0/30 is subnetted, 1 subnets
D   15.0.0.0/30 [90/3072] via 14.0.0.2, 00:04:26, GigabitEthernet0/0
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.16.0.0/16 is directly connected, GigabitEthernet0/1
L   172.16.0.1/32 is directly connected, GigabitEthernet0/1
D   173.16.0.0/16 [90/3328] via 14.0.0.2, 00:04:15, GigabitEthernet0/0

KandyNSBM#show ip protocols

Routing Protocol is "eigrp 100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 14.0.0.1
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
    14.0.0.0/30
    172.16.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    14.0.0.2         90           975457
  Distance: internal 90 external 170

KandyNSBM#
KandyNSBM#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address           Interface      Hold Uptime    SRTT   RTO   Q   Seq
                               (sec)          (ms)         Cnt   Num
0   14.0.0.2           Gig0/0        10  00:04:57    40    1000   0    5

KandyNSBM#

```

CopyPaste

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ColomboNSBM Routing Table

ColomboNSBM

Physical Config CLI Attributes

ColomboNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 14.0.0.0/30 is directly connected, GigabitEthernet0/0
L 14.0.0.2/32 is directly connected, GigabitEthernet0/0
15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 15.0.0.0/30 is directly connected, GigabitEthernet0/1
L 15.0.0.1/32 is directly connected, GigabitEthernet0/1
D 172.16.0.0/16 [90/3072] via 14.0.0.1, 00:06:11, GigabitEthernet0/0
D 173.16.0.0/16 [90/3072] via 15.0.0.2, 00:06:00, GigabitEthernet0/1

ColomboNSBM#show ip protocols

Routing Protocol is "eigrp 100"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Default networks flagged in outgoing updates
Default networks accepted from incoming updates
Redistributing: eigrp 100
EIGRP-IPv4 Protocol for AS(100)
Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
NSF-aware route hold timer is 240
Router-ID: 14.0.0.2
Topology: 0 (base)
Active Timer: 3 min
Distance: internal 90 external 170
Maximum path: 4
Maximum hopcount 100
Maximum metric variance 1

Automatic Summarization: disabled
Automatic address summarization:
Maximum path: 4
Routing for Networks:
14.0.0.0/30
15.0.0.0/30
Routing Information Sources:
Gateway Distance Last Update
14.0.0.1 90 971224
15.0.0.2 90 982644
Distance: internal 90 external 170

ColomboNSBM#
ColomboNSBM#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

Copy Paste

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ColomboNSBM EIGRP Neighbors

ColomboNSBM

Physical
Config
CLI
Attributes

```

- candidate default, 0 = per-user static route, 0 = ODR
P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
    C    14.0.0.0/30 is directly connected, GigabitEthernet0/0
    L    14.0.0.2/32 is directly connected, GigabitEthernet0/0
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
    C    15.0.0.0/30 is directly connected, GigabitEthernet0/1
    L    15.0.0.1/32 is directly connected, GigabitEthernet0/1
    D    172.16.0.0/16 [90/3072] via 14.0.0.1, 00:06:11, GigabitEthernet0/0
    D    173.16.0.0/16 [90/3072] via 15.0.0.2, 00:06:00, GigabitEthernet0/1

ColomboNSBM#show ip protocols

Routing Protocol is "eigrp 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 14.0.0.2
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
    14.0.0.0/30
    15.0.0.0/30
  Routing Information Sources:
    Gateway         Distance         Last Update
    14.0.0.1         90               971224
    15.0.0.2         90               982644
  Distance: internal 90 external 170

ColomboNSBM#
ColomboNSBM#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address          Interface         Hold Uptime      SRTT   RTO   Q   Seq
                               (sec)            (ms)          Cnt   Num
0   14.0.0.1          Gig0/0            13  00:06:23    40   1000   0    5
1   15.0.0.2          Gig0/1            10  00:06:12    40   1000   0    5

ColomboNSBM#

```

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GalleNSBM Routing Table

GalleNSBM

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```

GalleNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/30 is subnetted, 1 subnets
D       14.0.0.0/30 [90/3072] via 15.0.0.1, 00:07:59, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/1
L       15.0.0.2/32 is directly connected, GigabitEthernet0/1
D       172.16.0.0/16 [90/3328] via 15.0.0.1, 00:07:59, GigabitEthernet0/1
    173.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       173.16.0.0/16 is directly connected, GigabitEthernet0/2
L       173.16.0.1/32 is directly connected, GigabitEthernet0/2

GalleNSBM#show ip protocols

Routing Protocol is "eigrp 100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 15.0.0.2
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
    15.0.0.0/30
    173.16.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    15.0.0.1         90           979908
  Distance: internal 90 external 170

GalleNSBM#
GalleNSBM#

```

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GalleNSBM EIGRP Neighbors

GalleNSBM

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```

* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/30 is subnetted, 1 subnets
D    14.0.0.0/30 [90/3072] via 15.0.0.1, 00:07:59, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/30 is directly connected, GigabitEthernet0/1
L    15.0.0.2/32 is directly connected, GigabitEthernet0/1
D    172.16.0.0/16 [90/3328] via 15.0.0.1, 00:07:59, GigabitEthernet0/1
C    173.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    173.16.0.0/16 is directly connected, GigabitEthernet0/2
L    173.16.0.1/32 is directly connected, GigabitEthernet0/2

GalleNSBM#show ip protocols

Routing Protocol is "eigrp 100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 15.0.0.2
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
    15.0.0.0/30
    173.16.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    15.0.0.1         90            979908
  Distance: internal 90 external 170

GalleNSBM#
GalleNSBM#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address           Interface      Hold Uptime    SRTT   RTO   Q   Seq
  Address           Interface      (sec)  (sec)         (ms)          Cnt  Num
0   15.0.0.1           Gig0/1         13     00:08:30     40     1000   0    6

GalleNSBM#

```

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Successful Ping Test

EIGRP established neighbor relationships successfully and dynamically exchanged routes among all routers. Connectivity tests confirmed successful communication between all networks.

Conclusion

This practical assignment provided hands-on experience with Cisco router configuration and routing technologies. Basic router security was implemented through hostname configuration and console authentication. Static routing was configured manually to establish communication among three separate networks. Dynamic routing was then implemented using RIP and EIGRP protocols. Connectivity was verified through routing tables and successful ping tests. The assignment demonstrated the differences between static and dynamic routing and highlighted the efficiency of EIGRP in large-scale network environments.

GitHub Repository

Repository Name: **CNRoutingAssignment**

Repository Contents:

- Task 1 Packet Tracer File
- Task 2 Packet Tracer File
- Task 3 Packet Tracer File
- README.md Description
- Assignment Report PDF

GitHub Link: <https://github.com/mandfernando/CNRoutingAssignment>